Validation Determination Conflict & Uncertainty Governance Module (VDCUGM)

Architecture Ecosystem: Structured Reality Standards™
Architecture Family: VALIDOS® — Validation Governance Architecture
Parent Standard: Validation Determination Standard (VDTS)
Operational Layer: Validation Determination Governance Layer
Category: AI & Interpretation
Subcategory: Validation Governance
Type: Parent Standard Module
Governed Space: Validation Determination Conflict & Uncertainty Governance
Version: 1.0
Status: Canonical · Open Module
Effective Date: 11 August 2026
Compatibility: OOF® Methodology OS™ · GOA™ · OBIDENITY® · INTEGROS® · ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Governed Space

Validation Determination Conflict & Uncertainty Governance

Canonical Definition

Validation Determination Conflict & Uncertainty Governance Module (VDCUGM) defines the governance framework for identifying, classifying, preserving, analyzing, contextualizing, escalating, constraining, and where methodologically possible resolving material conflicts, contradictory findings, incompatible criterion signals, uncertainty, ambiguity, incomplete knowledge, and determination instability arising within the governed Validation Determination Basis.

It establishes the conditions under which disagreement and uncertainty become explicit determination-governance inputs rather than hidden, averaged, discarded, or prematurely converted into certainty.

Operational Role

VDCUGM serves as the third internal governance space of the Validation Determination Standard.

The preceding modules establish:

VDBRQM — Which results may legitimately participate in determination?

VCDM — What does the qualified result set demonstrate about each individual Validation Criterion?

VDCUGM asks:

Where do legitimate findings conflict, where does uncertainty materially affect interpretation, and what must happen before criterion outcomes can be integrated into an overall Validation Determination?

The governed progression is:

Criterion Determinations & Conflict Signals → Conflict Identification → Conflict Classification → Uncertainty Identification → Materiality Assessment → Cause & Scope Analysis → Resolution or Preservation → Determination Constraint → Conflict & Uncertainty Record

VDCUGM does not perform final cross-criterion aggregation.

It governs the interpretive instability that must be understood before aggregation can legitimately occur.

Module Operational Space

VDCUGM governs:

- determination conflict,
- criterion conflict,
- result conflict,
- evidence conflict,
- source-related determination conflict,
- method-related determination conflict,
- execution-related determination conflict,
- scope conflict,
- threshold conflict,
- temporal conflict,
- contextual conflict,
- contradiction,
- inconsistency,
- unresolved disagreement,
- determination uncertainty,
- measurement uncertainty,
- methodological uncertainty,
- evidentiary uncertainty,
- source uncertainty,
- execution uncertainty,
- sampling uncertainty,

- contextual uncertainty,
- model uncertainty,
- object-state uncertainty,
- uncertainty materiality,
- uncertainty propagation,
- conflict materiality,
- conflict causation,
- conflict localization,
- conflict escalation,
- conflict resolution,
- unresolved conflict preservation,
- conditional determination constraints,
- indeterminacy triggers,
- and conflict-and-uncertainty traceability.

Module Function

VDCUGM applies after criterion-level determinations and material conflict signals have been established.

Its function is to prevent:

- contradictory findings being silently averaged,
- unfavorable evidence being discarded as noise,
- uncertainty being hidden behind point estimates,
- disagreements between methods being resolved by preference,
- disagreements between sources being resolved by reputation alone,
- scope differences being mistaken for factual contradiction,
- temporal changes being treated as inconsistent evidence,
- missing information being converted into certainty,
- one dominant criterion result suppressing legitimate contrary evidence,
- and overall determination being constructed before material conflict and uncertainty are governably understood.

Conflict Is Not Automatically Error

VALIDOS® establishes:

Conflict ≠ Methodological Failure

Two legitimate results may disagree because they describe:

- different operating conditions,
- different populations,
- different times,
- different system states,
- different methodological sensitivities,
- or genuinely unstable object behavior.

Conflict may therefore contain important validation information.

The first governance task is not to eliminate conflict.

It is to understand what the conflict represents.

Validation Determination Conflict

A **Validation Determination Conflict** exists where two or more legitimate findings, criterion outcomes, or interpretive pathways support materially incompatible conclusions within an overlapping determination space.

The overlap requirement matters.

Two different conclusions are not necessarily conflicting if they apply to different valid scopes.

Result Conflict

A Result Conflict exists where qualified Validation Results materially disagree about the same validation-relevant condition.

Examples include:

- repeated measurements producing incompatible values,
- two tests producing different classifications,
- simulation and operational testing diverging,
- or different time periods showing materially different behavior.

Criterion Conflict

A Criterion Conflict exists where one criterion-level determination contains legitimate internal inconsistency or where materially related criteria support incompatible implications.

For example:

Criterion A: Satisfied

while:

Criterion B: Not Satisfied

may or may not constitute a conflict.

It becomes a determination conflict only if the approved criteria structure expects those outcomes to be mutually compatible or causally related.

Evidence Conflict

Evidence Conflict occurs where legitimate evidence supports materially different interpretations.

This may arise from:

- different datasets,
- different populations,
- different observations,
- different time periods,
- or different environments.

Evidence Conflict is not automatically Source Conflict.

Source-Related Determination Conflict

Two sufficiently reliable sources may disagree.

VDCUGM preserves the disagreement and examines whether it arises from:

- different source roles,
- different observations,
- different access,
- different time periods,
- different methodologies,
- or genuine factual incompatibility.

A more authoritative source should not automatically win merely because of status.

Method-Related Determination Conflict

Different suitable Validation Methods may produce materially different findings.

This may reveal:

- different sensitivity,
- different assumptions,
- different scope,
- hidden methodological limitations,
- or object behavior that varies by measurement method.

Method conflict may require methodological comparison rather than numerical averaging.

Execution-Related Conflict

Execution differences may explain apparent determination conflict.

For example:

Two valid test results may differ because:

- one occurred under nominal operation,
- one under degraded operation,
- one used a different configuration,
- or one followed a different sequence.

VDCUGM distinguishes factual disagreement from execution-context difference.

Scope Conflict

A Scope Conflict exists where two determinations appear contradictory but actually apply to materially different scopes.

For example:

Satisfied for Population A

and:

Not Satisfied for Population B

are not necessarily contradictory.

The conflict may disappear once scope is correctly represented.

Temporal Conflict

A Validation Object may change over time.

A criterion may be satisfied at Time 1 and not satisfied at Time 2.

This is not necessarily inconsistent validation.

It may represent:

State Change

Drift

Degradation

Improvement

or:

Context Change

VDCUGM preserves time as part of

conflict interpretation.

Contextual Conflict

A system may behave differently across:

- jurisdictions,
- languages,
- environments,
- user groups,
- hardware,
- software configurations,
- or operating modes.

Contextual differences must not be compressed into one universal conclusion if the underlying behavior is genuinely conditional.

Threshold Conflict

Different thresholds may produce apparently conflicting conclusions.

VDCUGM identifies whether:

- different criterion versions were used,
- different jurisdictions apply different thresholds,
- different Validation Depths use different acceptance levels,
- or an unauthorized threshold change occurred.

Threshold differences must remain explicit.

Conflict Classification

A material conflict may be classified as:

Apparent Conflict

The disagreement disappears when scope, time, context, or definition is clarified.

Resolvable Conflict

The conflict can be methodologically resolved using existing governed information.

Conditionally Resolvable Conflict

The conflict can be resolved only under explicit assumptions, boundaries, or additional controls.

Unresolved Conflict

Legitimate incompatible findings remain after reasonable governance analysis.

Critical Conflict

The unresolved conflict materially prevents a reliable criterion or overall determination.

These classifications support downstream determination governance.

Conflict Materiality

Not every disagreement matters equally.

Conflict materiality may depend upon:

- affected criterion,
- mandatory or critical status,
- Validation Purpose,
- Validation Depth,
- scope,
- consequence of error,
- persistence,
- strength of competing findings,
- and whether the conflict can change the overall determination.

Conflict Localization

VDCUGM seeks to determine where the conflict actually exists.

Possible localization levels include:

- one result,
- one criterion,
- one subgroup,
- one environment,
- one time period,
- one subsystem,
- one object state,
- or the complete Validation Object.

Localization prevents narrow conflicts from unnecessarily contaminating the entire determination.

Conflict Cause Analysis

Possible causes may include:

- measurement variation,

- evidence limitations,
- source differences,
- method differences,
- execution differences,
- sampling effects,
- data drift,
- object instability,
- context dependence,
- threshold mismatch,
- hidden dependency,
- or genuine unresolved reality.

Cause analysis informs whether conflict can be resolved, bounded, or must remain explicit.

Conflict Resolution

Where methodologically justified, conflict may be resolved through:

- scope clarification,
- corrected threshold application,
- removal of ineligible result,
- corrected result qualification,
- method comparison,
- additional analysis,
- additional validation,
- source clarification,
- evidence clarification,
- temporal separation,
- contextual separation,
- or another governed mechanism.

Resolution Must Not Mean Preference

VALIDOS® establishes:

Conflict Resolution ≠ Choosing the Preferred Result

Resolution must be based upon governed methodological reasons.

A favorable conclusion does not gain priority merely because it supports deployment, approval, reputation, or business objectives.

Unresolved Conflict

An unresolved conflict is a legitimate determination condition.

Possible downstream effects include:

- Criterion Not Determinable,

- Conditional Determination,
- Partial Determination,
- reduced determination scope,
- additional validation requirement,
- or Indeterminate Validation Determination.

Conflict Preservation

Where conflict cannot be legitimately resolved, both sides of the conflict should remain traceable.

The record should preserve:

- conflicting findings,
- their basis,
- relevant scope,
- evidence,
- sources,
- methods,
- uncertainty,
- and reason resolution was not possible.

Uncertainty Governance

Validation uncertainty represents the degree to which available governed information leaves material doubt concerning interpretation or determination.

Uncertainty is not one single phenomenon.

VDCUGM separates uncertainty by source and function.

Measurement Uncertainty

Measurement uncertainty concerns uncertainty inherent in observed or measured values.

This may include:

- instrument tolerance,
- noise,
- repeatability,
- precision,
- resolution,
- or another measurement limitation.

Methodological Uncertainty

Methodological uncertainty concerns uncertainty arising from:

- method assumptions,
- model choice,
- analytical design,
- simulation limitations,
- sampling method,
- or methodological approximation.

Evidentiary Uncertainty

Evidentiary uncertainty concerns:

- incomplete evidence,
- limited representativeness,
- sparse observations,
- missing scenarios,
- temporal limitations,
- or Evidence Fitness restrictions.

Source Uncertainty

Source uncertainty concerns unresolved conditions involving:

- identity,
- provenance,
- capability,
- competence,
- independence,
- conflicts,
- or reliability.

Execution Uncertainty

Execution uncertainty concerns uncertainty about:

- what happened,
- whether a procedure was followed,
- whether a control remained active,
- whether object state changed,
- or whether critical execution details can be reconstructed.

Sampling Uncertainty

Sampling may produce uncertainty concerning whether observed results represent the wider population, environment, or state space.

Contextual Uncertainty

Contextual uncertainty arises where applicability across:

- environments,
- populations,
- jurisdictions,
- languages,
- system states,
- or operating conditions

cannot be fully established.

Object-State Uncertainty

The Validation Object itself may exhibit:

- stochastic behavior,
- adaptive behavior,
- dynamic behavior,
- drift,
- emergent behavior,
- or poorly understood state transitions.

This may create uncertainty that is not reducible to measurement error.

Uncertainty Is Not Ignorance

Some uncertainty can be characterized quantitatively or qualitatively.

Other uncertainty reflects lack of knowledge.

VDCUGM permits distinctions such as:

Known Quantified Uncertainty

Known Qualitative Uncertainty

Known Unknown

and:

Uncharacterized Uncertainty

where useful.

Uncertainty Materiality

Uncertainty becomes material where it can reasonably affect:

- criterion status,
- determination scope,
- determination conditions,
- overall Validation Determination,
- or later reliance.

Not every uncertainty requires the same governance response.

Uncertainty Tolerance

The acceptable degree of uncertainty may depend upon:

- Validation Purpose,
- Validation Depth,
- criterion criticality,
- risk,
- reversibility,
- consequence of error,
- and intended reliance.

A research context may tolerate uncertainty unacceptable in high-consequence autonomous deployment.

Uncertainty-Compatible Determination

A determination may remain legitimate where uncertainty is sufficiently bounded and compatible with the applicable governance requirements.

Perfect certainty is not required.

Uncertainty-Driven Conditionality

Where uncertainty restricts applicability but does not prevent a conclusion, downstream determination may become conditional.

For example:

Positive only within tested operating conditions.

Uncertainty-Driven Indeterminacy

Where uncertainty prevents legitimate distinction between satisfaction and non-satisfaction, the correct outcome is indeterminacy.

VALIDOS® establishes:

When uncertainty materially prevents knowing, the architecture should preserve not knowing.

Uncertainty Propagation

Uncertainty should remain attached to the findings that depend upon it.

The conceptual chain may be:

Evidence Uncertainty → Result Uncertainty → Criterion Uncertainty → Overall Determination Constraint

This prevents uncertainty from disappearing during aggregation.

Uncertainty Amplification

Several individually modest uncertainties may combine into a larger determination problem.

For example:

- moderate evidence uncertainty,
- moderate source uncertainty,
- and moderate execution uncertainty

may collectively create substantial criterion uncertainty.

VDCUGM therefore considers cumulative uncertainty.

Uncertainty Reduction

Uncertainty may be reduced through:

- additional evidence,
- additional validation,
- better measurement,
- independent corroboration,
- narrower scope,
- improved source qualification,
- repeated testing,
- or clearer methodological boundaries.

Uncertainty reduction should remain traceable.

Residual Uncertainty

Uncertainty remaining after reasonable reduction efforts becomes
Residual Uncertainty.

Residual Uncertainty must propagate into later determination governance where material.

Conflict and Uncertainty Interaction

Conflict and uncertainty often interact.

A conflict may increase uncertainty.

Uncertainty may make apparent conflict impossible to resolve.

VDCUGM therefore governs them together while preserving their conceptual distinction.

Conflict Without High Uncertainty

Two strong findings may genuinely conflict.

In such cases, the conflict may be clear even though each finding is individually well-supported.

High Uncertainty Without Conflict

A single criterion may have no contradictory result but still remain highly uncertain because the evidence basis is weak.

Therefore:

Conflict ≠ Uncertainty

and: **Uncertainty ≠ Conflict**

Determination Constraint

VDCUGM may create a formal **Determination Constraint.**

A Determination Constraint defines how identified conflict or uncertainty must affect later cross-criterion integration.

Possible constraints include:

- reduced scope,
- conditional interpretation,
- no aggregation across specified results,

- additional validation required,
- criterion remains Not Determinable,
- partial determination only,
- or overall indeterminacy required.

Critical Determination Constraint

A critical conflict or uncertainty may prevent legitimate overall determination.

Such a condition may create:

Overall Determination Blocked by Conflict

or:

Overall Determination Blocked by Uncertainty

where appropriate.

Conflict Escalation

Material conflicts may require escalation where resolution exceeds the authority or capability of the current determination process.

Escalation may involve:

- methodology review,
- independent review,
- specialized expert review,
- governance authority,
- additional validation,
- or another defined pathway.

Uncertainty Escalation

High-consequence uncertainty may likewise require:

- deeper Validation Depth,
- additional evidence,
- broader testing,
- more reliable sources,
- or independent confirmation.

Conflict Status

VDCUGM may establish:

Conflict Resolved

Conflict Conditionally Resolved

Conflict Bounded

Conflict Unresolved

Critical Conflict

or:

Conflict Status Undetermined

Uncertainty Status

Uncertainty may be characterized as:

Uncertainty Acceptable

Uncertainty Conditionally Acceptable

Uncertainty Material

Uncertainty Determination-Limiting

Uncertainty Determination-Blocking

or:

Uncertainty Undetermined

These statuses inform later integration rather than becoming Validation States.

Conflict & Uncertainty Record

A formal record may preserve:

- Conflict or Uncertainty Identifier,
- affected Validation Object,
- affected criteria,
- affected results,
- scope,
- time,
- context,
- evidence,
- sources,
- methods,
- execution references,
- materiality,

- cause,
- uncertainty type,
- competing interpretations,
- analysis,
- resolution attempt,
- resolution status,
- residual uncertainty,
- Determination Constraints,
- escalation,
- and sufficient traceability.

Human Conflict Resolution

Human expert judgment may assist conflict resolution where interpretation is complex.

Where used, it should remain:

- attributable,
- bounded,
- competence-aware,
- conflict-of-interest aware,
- evidence-based,
- and traceable.

Automated Conflict Detection

Automated systems may identify:

- inconsistent results,
- threshold disagreement,
- cross-method divergence,
- duplicate contradictions,
- or statistical anomalies.

Detection can be automated even where resolution requires human or governed methodological review.

AI-Assisted Conflict & Uncertainty Analysis

AI may assist with:

- clustering conflicting findings,
- tracing sources,
- identifying contextual differences,
- mapping uncertainty,
- detecting hidden assumptions,
- and proposing resolution pathways.

AI assistance should not silently resolve a material conflict by preference or opaque reasoning.

Conflict Reassessment

Conflict status may change when:

- new results appear,
- evidence is corrected,
- scope is clarified,
- source reliability changes,
- a method error is identified,
- or additional validation is performed.

Uncertainty Reassessment

Uncertainty may change because of:

- new evidence,
- additional observation,
- improved measurement,
- model change,
- environmental change,
- or improved methodological understanding.

Historical uncertainty assessments should remain visible.

Minimum Implementation Framework

1. Identify Material Conflicts

Detect incompatible findings, criterion signals, or interpretations within overlapping determination scope.

2. Classify Conflict Type

Determine whether conflict is:

- result-based,
- criterion-based,
- evidence-based,
- source-related,
- method-related,
- execution-related,
- scope-related,
- temporal,
- contextual,

- or threshold-related.

3. Identify Uncertainty

Map relevant uncertainty sources affecting criterion or overall interpretation.

4. Assess Materiality

Determine whether conflict or uncertainty is:

- immaterial,
- tolerable,
- material,
- determination-limiting,
- or determination-blocking.

5. Analyze Cause and Scope

Determine whether the issue reflects:

- actual contradiction,
- scope difference,
- time difference,
- context difference,
- methodological limitation,
- source difference,
- execution difference,
- or genuine object instability.

6. Attempt Governed Resolution Where Legitimate

Use methodologically defensible resolution pathways.

7. Preserve Unresolved Conflict and Residual Uncertainty

Do not force closure where the governed basis does not support it.

8. Establish Determination Constraints

Define how conflict and uncertainty must affect later cross-criterion integration.

9. Establish Escalation Where Required

Trigger additional review or validation where materiality demands it.

10. Preserve Lifecycle Traceability

Maintain:

- conflict records,
- uncertainty records,
- affected criteria,
- affected results,
- evidence,
- sources,
- methods,
- execution references,
- scope,
- context,
- time,
- materiality,
- cause,
- analysis,
- resolution status,
- residual uncertainty,
- Determination Constraints,
- reassessment triggers,
- changes,
- and sufficient lifecycle continuity.

Governance Outputs

VDCUGM may produce:

- Validation Determination Conflict Register,
- Result Conflict Record,
- Criterion Conflict Record,
- Evidence Conflict Record,
- Source-Related Conflict Record,
- Method-Related Conflict Record,
- Execution-Related Conflict Record,
- Scope Conflict Record,
- Temporal Conflict Record,
- Contextual Conflict Record,
- Threshold Conflict Record,
- Conflict Classification,
- Conflict Materiality Assessment,
- Conflict Cause Assessment,
- Conflict Localization Record,
- Conflict Resolution Record,
- Unresolved Conflict Record,
- Validation Uncertainty Register,
- Measurement Uncertainty Record,
- Methodological Uncertainty Record,
- Evidentiary Uncertainty Record,
- Source Uncertainty Record,
- Execution Uncertainty Record,
- Sampling Uncertainty Record,
- Contextual Uncertainty Record,
- Object-State Uncertainty Record,
- Residual Uncertainty Record,
- Cumulative Uncertainty Assessment,
- Determination Constraint,

- Critical Determination Constraint,
- Conflict Escalation Record,
- Uncertainty Escalation Record,
- Conflict Status,
- Uncertainty Status,
- and Conflict & Uncertainty Reassessment Record.

Use Case 1 — AI Model Validation

Scenario

An AI model satisfies a safety criterion in simulation but produces materially different behavior during controlled real-world testing.

Both validation methods were suitable and both executions were conformant.

Application

VDCUGM does not select the more favorable result.

It examines:

- method differences,
- environmental differences,
- scope,
- execution conditions,
- object state,
- and remaining uncertainty.

Result

The conflict may be localized to simulation-to-reality transfer and produce a Determination Constraint preventing unrestricted criterion satisfaction until the divergence is resolved or bounded.

Use Case 2 — Healthcare AI

Scenario

A diagnostic AI system meets the required performance threshold across the overall validation population but shows materially lower performance in a small subgroup with limited sample size.

Application

VDCUGM distinguishes:

- subgroup conflict,
- sampling uncertainty,
- population scope,

- and evidence limitations.

Result

Rather than averaging the subgroup away, the module may preserve conditionality or require the subgroup criterion to remain Not Determinable pending additional validation.

Use Case 3 — Autonomous System

Scenario

Repeated validation of an autonomous system produces stable results under normal conditions but inconsistent outcomes under a rare degraded-state scenario.

Application

VDCUGM determines whether the inconsistency reflects:

- stochastic system behavior,
- insufficient sample size,
- environment variation,
- execution differences,
- or genuine unstable behavior.

Result

Where the conflict cannot be resolved, the residual uncertainty remains attached to the affected criterion and may constrain later overall determination.

Architectural Position

Validation Determination Conflict & Uncertainty Governance is the third internal governance space of the **Validation Determination Standard (VDTS).**

The progression now becomes:

Determination Basis & Result Qualification → Criterion Determination → Conflict & Uncertainty Governance

VDBRQM establishes:

Which results may participate?

VCDM establishes:

What does each qualified result set demonstrate about

each criterion?

VDCUGM establishes:

Where do those findings conflict, where does uncertainty remain, and what interpretive constraints must survive?

The next VDTS governance stage can then ask:

How should the now-governed criterion outcomes be combined across mandatory, critical, dependent, conditional, and hierarchical criteria?

The VDTS progression therefore continues:

Determination Basis & Result Qualification → Criterion Determination → Conflict & Uncertainty Governance → Cross-Criterion Integration → Overall Validation Determination

---

Foundational Principle

Conflict and uncertainty should not be eliminated for the sake of a cleaner conclusion; they should remain visible until the validation record legitimately supports resolution, limitation, or indeterminacy.

Canonical Closing Statement

Validation Determination Conflict & Uncertainty Governance Module establishes the interpretive stability layer of VALIDOS® by governing contradictory findings, criterion conflict, evidence and source disagreement, method and execution divergence, scope and temporal differences, uncertainty, materiality, causation, localization, resolution, residual uncertainty, escalation, and Determination Constraints. VDCUGM ensures that cross-criterion integration cannot convert unresolved disagreement, methodological ambiguity, or incomplete knowledge into artificial certainty and that every later Validation Determination remains bounded by the conflicts and uncertainties the governed validation record is actually capable of resolving.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer]

[→ VALIDOS® Architecture Map]

→ VALIDOS® Complete Standards & Modules Index

Internal Module Flow

→ Previous module: Validation Criterion Determination Module (VCDM)

→ Next module: Validation Cross-Criterion Integration Module (VCCIM)

Related Documents

→ Validation Determination Standard

→ About Validation Determination Standard

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

https://originopenfoundation.org/